Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		$4CuFeS_2$ + **10½**O_2 → **4**Cu + **2**FeO + Fe_2O_3 + **8**SO_2		1		1		
		(ii)		$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$	1			1		
		(iii)		M_r $CuFeS_2$ = 183.5 (1) % Cu = $\frac{1.3}{100} \times \frac{63.5}{183.5} \times 100 = 0.45$ % (1)		2		2	1	
		(iv)		Cu reduced oxidation number changes from +1 to 0 (1) S oxidised oxidation number changes from –2 to +4 (1) O reduced oxidation number changes from 0 to –2 (1)		3		3		
	(b)			strong acid is one that fully dissociates in aqueous solution (1) concentrated acid consists of a large quantity of acid and a small quantity of water (1)	2			2		
	(c)	(i)		moles NaOH = 0.0125 × 0.0159 = 1.988×10^{-4} (1) concentration acid = $\frac{1.988 \times 10^{-4}}{0.025} = 7.95 \times 10^{-3}$ mol dm^{-3} (1)		2		2	1	2
		(ii)		if pH = 2.10 then $[H^+] = 7.94 \times 10^{-3}$ mol dm^{-3} (1) since this is the same as the concentration of the acid, it must have fully dissociated (and teacher is correct) (1)			2	2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		colour change from yellow-green to blue (1) the system will try to minimise this effect by using up the water and so the position of equilibrium moves to the left (1)			2	2		2
		(ii)		colour changes from blue to yellow-green (1) the system opposes the change by taking in heat so the position of equilibrium moves in the endothermic direction (1)			2	2		2
				Question 8 total	**3**	**8**	**6**	**17**	**3**	**6**